

Roll No.:

MID SEMESTER EXAMINATION SEPT. 2025

Semester: V

Name of the Program: **B.Tech.**

Name of the Course: Cloud-Based Application Development and Management

Course Code: TCS 552

Maximum Marks: 50

Time : 90 Minutes

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each question carries 10 marks

	(10 Marks)	CO1
(a)	Identify the critical components of a modern cloud data center and discuss their role in service provisioning.	
	OR	
(b)	Critically evaluate the impact of cloud adoption on enterprise IT governance, cost optimization, and innovation.	
Q2	(10 Marks)	CO3
(a)	Examine the concept of virtualization and discuss how it enhances resource efficiency in cloud environments. Outline the types of virtualization.	
	OR	
(b)	Explain how virtualization enables efficient resource utilization in cloud environments. Discuss BareMetal and Hosted Hypervisor.	
Q3	(10 Marks)	CO3
(a)	Analyze the significance of Service Level Agreements (SLAs) in maintaining trust and accountability in cloud services.	
	OR	
(b)	Explain the role of middleware and software stacks in enabling cloud-based applications.	
Q4	(10 Marks)	CO2
(a)	Examine the architecture of Google App Engine and analyze its application life cycle with reference to real-world scenarios.	
	OR	
(b)	Explain the role of Heroku in modern application deployment.	
Q5	(10 Marks)	CO1
(a)	List the core compute services offered by Amazon Web Services (AWS).	
	OR	
(b)	Evaluate the advantages and limitations of IBM Cloud (Kyndryl).	CO2

